The Hong Kong University of Science and Technology
IEDA 1250, Summer 2026
Final Exam Solution

Student ID ____________________ Name: ________________________________

Please read the following instructions carefully.

- You have **120** minutes to complete this exam. This question booklet contains **7** questions for a total of 120 points. Check to see if any pages are missing.
- Read the instructions for individual questions carefully before answering the questions.
- **Please write your answers in the answer book.** You can use this question booklet as scratch paper.

Question	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
6	10	
7	10	
Total	120	

1. (20 points) Consider the zero-sum game having the following payoff table for Player 1:

		Player 2	
		S_1	S_2
Player 1	S_1	3	-2
	S_2	-1	2

(a) Use the graphical procedure to determine the value of the game and the optimal mixed strategy for each player.

(b) Check your answer by (i) constructing Player 2's payoff table and (ii) applying the graphical procedure again to the resulting table.

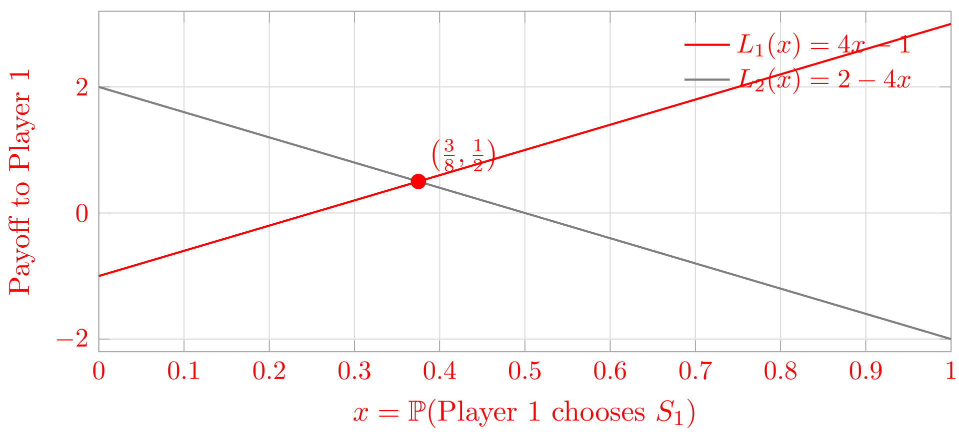
Solution.

(a) Let Player 1 choose S_1 with probability x and S_2 with probability $1-x$. If Player 2 chooses S_1 or S_2, respectively, Player 1's expected payoff is

$$L_1(x) = 3x - (1-x) = 4x - 1, \qquad L_2(x) = -2x + 2(1-x) = 2 - 4x.$$

Player 1 maximizes the lower envelope of these two lines. Graphically, the maximum occurs at their intersection:

$$4x - 1 = 2 - 4x \quad \Longrightarrow \quad x = \frac{3}{8}, \qquad v = \frac{1}{2}.$$



Therefore, Player 1's optimal strategy is

$$p^* = \left(\frac{3}{8}, \frac{5}{8}\right).$$

Let Player 2 choose S_1 with probability q. To make Player 1 indifferent between the two rows,

$$3q - 2(1-q) = -q + 2(1-q) \quad \Longrightarrow \quad q = \frac{1}{2}.$$

Hence Player 2's optimal strategy and the value to Player 1 are

$$q^* = \left(\frac{1}{2}, \frac{1}{2}\right), \qquad v = \frac{1}{2}.$$

(b) Player 2's payoff is the negative of Player 1's payoff. With Player 2's strategies as rows, its payoff table is

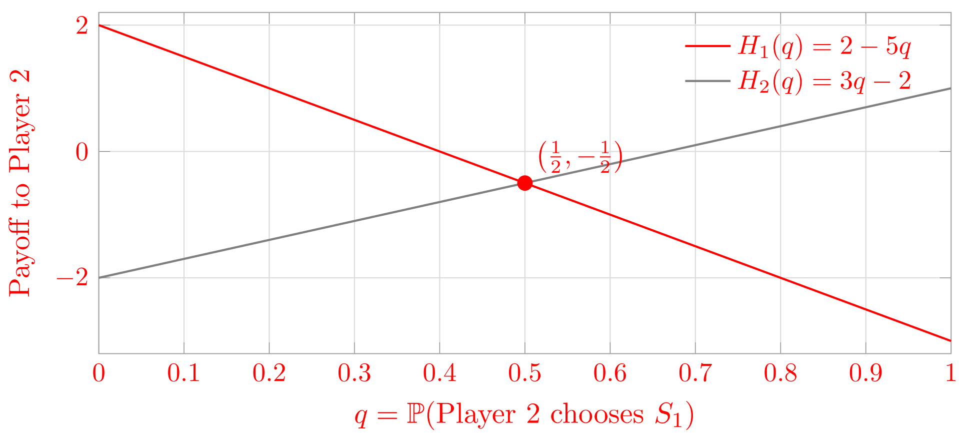
	Player 1 S_1	Player 1 S_2
Player 2 S_1	-3	1
Player 2 S_2	2	-2

If Player 2 chooses its first row with probability q, its payoff against Player 1's two columns is

$$H_1(q) = -3q + 2(1-q) = 2 - 5q, \qquad H_2(q) = q - 2(1-q) = 3q - 2.$$

The lower-envelope maximum satisfies

$$2 - 5q = 3q - 2 \quad \Longrightarrow \quad q = \frac{1}{2}, \qquad v_2 = -\frac{1}{2}.$$



Finally, Player 1 makes Player 2 indifferent between its rows:

$$-3x + (1-x) = 2x - 2(1-x) \quad \Longrightarrow \quad x = \frac{3}{8}.$$

Thus the second graphical calculation confirms

$$\boxed{p^* = \left(\frac{3}{8}, \frac{5}{8}\right), \qquad q^* = \left(\frac{1}{2}, \frac{1}{2}\right), \qquad v_1 = \frac{1}{2}, \quad v_2 = -\frac{1}{2}}$$

2. (20 points) Consider the following zero-sum two-player game. The entries in the table are the payoffs to Player A.

	B_1	B_2	B_3	B_4	B_5	B_6
A_1	1	10	2	3	11	12
A_2	4	8	5	6	9	10
A_3	7	5	8	9	6	7
A_4	10	1	11	12	2	3
A_5	3	7	4	5	8	9
A_6	6	4	7	8	5	6

Player A chooses one of the six rows, and Player B chooses one of the six columns. Player A wants to maximize the expected payoff, while Player B wants to minimize the expected payoff.

(a) Find the Nash equilibrium in this game and report the value of the game.

(b) Write Player A's linear program for the *full* payoff table.

(c) The following linear program is associated with Player B. Let q_j be the probability that Player B chooses column j, for $j = 1, \ldots, 6$, and let v_B represent the expected loss of Player B:

$$\begin{aligned}
\min \quad & v_B \\
\text{s.t.} \quad & q_1 + 10q_2 + 2q_3 + 3q_4 + 11q_5 + 12q_6 \le v_B, \\
& 4q_1 + 8q_2 + 5q_3 + 6q_4 + 9q_5 + 10q_6 \le v_B, \\
& 7q_1 + 5q_2 + 8q_3 + 9q_4 + 6q_5 + 7q_6 \le v_B, \\
& 10q_1 + q_2 + 11q_3 + 12q_4 + 2q_5 + 3q_6 \le v_B, \\
& 3q_1 + 7q_2 + 4q_3 + 5q_4 + 8q_5 + 9q_6 \le v_B, \\
& 6q_1 + 4q_2 + 7q_3 + 8q_4 + 5q_5 + 6q_6 \le v_B, \\
& q_1 + q_2 + q_3 + q_4 + q_5 + q_6 = 1, \\
& q_1, q_2, q_3, q_4, q_5, q_6 \ge 0, \\
& v_B \text{ is unrestricted in sign.}
\end{aligned}$$

Show that Player A's LP and Player B's LP are primal and dual LPs. That is, starting from Player A's LP in part (b), derive its dual and show that the dual is identical to the given one.

Solution.

(a) First eliminate dominated strategies. For Player A, row A_5 is strictly dominated by A_2, and row A_6 is strictly dominated by A_3. For Player B, columns B_3 and B_4 are dominated by B_1, and columns B_5 and B_6 are dominated by B_2. The reduced payoff matrix is

	B_1	B_2
A_1	1	10
A_2	4	8
A_3	7	5
A_4	10	1

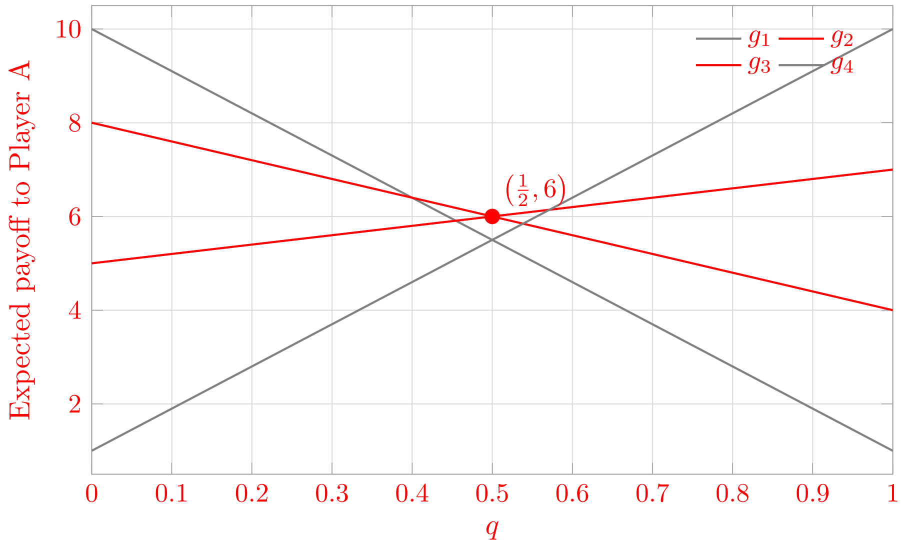
.

Let q be the probability that Player B chooses B_1. The expected payoff to Player A from each remaining row is

$$g_1(q) = 10 - 9q, \quad g_2(q) = 8 - 4q,$$
$$g_3(q) = 5 + 2q, \quad g_4(q) = 1 + 9q.$$

Player B minimizes the upper envelope. The relevant intersection is between g_2 and g_3:

$$8 - 4q = 5 + 2q \quad \Longrightarrow \quad q = \frac{1}{2}, \qquad v = 6.$$



Thus Player B's optimal strategy is

$$q_1 = q_2 = \frac{1}{2}, \qquad q_3 = q_4 = q_5 = q_6 = 0.$$

To find Player A's strategy, let Player A choose A_2 with probability p and A_3 with probability $1-p$. Indifference between B_1 and B_2 gives

$$4p + 7(1-p) = 8p + 5(1-p) \quad \Longrightarrow \quad p = \frac{1}{3}.$$

Therefore,

$$\boxed{p^* = \left(0, \frac{1}{3}, \frac{2}{3}, 0, 0, 0\right), \qquad q^* = \left(\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right), \qquad v = 6}$$

As a check, p^* gives $(6, 6, 7, 8, 7, 8)$ against the six columns, while q^* gives $(11/2, 6, 6, 11/2, 5, 5)$ for the six rows.

(b) Let p_i be the probability that Player A chooses row i, and let v_A be Player A's guaranteed payoff. Player A's LP for the full table is

$$\begin{aligned}
\max \quad & v_A \\
\text{s.t.} \quad & p_1 + 4p_2 + 7p_3 + 10p_4 + 3p_5 + 6p_6 \geq v_A, \\
& 10p_1 + 8p_2 + 5p_3 + p_4 + 7p_5 + 4p_6 \geq v_A, \\
& 2p_1 + 5p_2 + 8p_3 + 11p_4 + 4p_5 + 7p_6 \geq v_A, \\
& 3p_1 + 6p_2 + 9p_3 + 12p_4 + 5p_5 + 8p_6 \geq v_A, \\
& 11p_1 + 9p_2 + 6p_3 + 2p_4 + 8p_5 + 5p_6 \geq v_A, \\
& 12p_1 + 10p_2 + 7p_3 + 3p_4 + 9p_5 + 6p_6 \geq v_A, \\
& p_1 + p_2 + p_3 + p_4 + p_5 + p_6 = 1, \\
& p_i \geq 0 \quad (i = 1, \ldots, 6), \qquad v_A \in \mathbb{R}.
\end{aligned}$$

(c) Rewrite the six payoff constraints as

$$v_A - \sum_{i=1}^{6} a_{ij} p_i \le 0, \qquad j = 1, \dots, 6.$$

Associate $q_j \ge 0$ with these six constraints and associate the unrestricted dual variable v_B with $\sum_i p_i = 1$. The dual objective is $\min v_B$. The variables $p_i \ge 0$ produce

$$v_B - \sum_{j=1}^{6} a_{ij} q_j \ge 0 \quad \Longleftrightarrow \quad \sum_{j=1}^{6} a_{ij} q_j \le v_B, \qquad i = 1, \dots, 6.$$

Because v_A is unrestricted, its dual constraint is

$$\sum_{j=1}^{6} q_j = 1.$$

Consequently, the dual is

$$\begin{aligned}
\min \quad & v_B \\
\text{s.t.} \quad & q_1 + 10q_2 + 2q_3 + 3q_4 + 11q_5 + 12q_6 \le v_B, \\
& 4q_1 + 8q_2 + 5q_3 + 6q_4 + 9q_5 + 10q_6 \le v_B, \\
& 7q_1 + 5q_2 + 8q_3 + 9q_4 + 6q_5 + 7q_6 \le v_B, \\
& 10q_1 + q_2 + 11q_3 + 12q_4 + 2q_5 + 3q_6 \le v_B, \\
& 3q_1 + 7q_2 + 4q_3 + 5q_4 + 8q_5 + 9q_6 \le v_B, \\
& 6q_1 + 4q_2 + 7q_3 + 8q_4 + 5q_5 + 6q_6 \le v_B, \\
& q_1 + q_2 + q_3 + q_4 + q_5 + q_6 = 1, \\
& q_j \ge 0 \quad (j = 1, \dots, 6), \qquad v_B \in \mathbb{R}.
\end{aligned}$$

This is identical to the given Player B LP, so the two LPs are primal and dual.

$$\boxed{\text{Player A's LP and the given Player B LP form a primal–dual pair.}}$$

3. (20 points) A college student has 7 days remaining before final examinations begin in her four courses, and she wants to allocate this study time as effectively as possible. She needs at least 1 day on each course, and she likes to concentrate on just one course each day, so she wants to allocate 1, 2, 3, or 4 days to each course. Having recently taken an optimization course, she decides to use dynamic programming to make these allocations to maximize the total grade points to be obtained from the four courses. She estimates that the alternative allocations for each course would yield the number of grade points shown in the following table:

Study Days	Course 1	Course 2	Course 3	Course 4
1	3	5	2	6
2	5	5	4	7
3	6	6	7	9
4	7	9	8	9

Solve this problem by dynamic programming. Show the backward induction calculations. You do not need to write down the recursive equations.

Solution.

Let $G_i(s)$ denote the maximum total grade points obtainable from Courses $i, i+1, \ldots, 4$ when exactly s study days remain.

For Course 4,

$$\begin{array}{c|cccc} s & 1 & 2 & 3 & 4 \\ \hline G_4(s) & 6 & 7 & 9 & 9 \end{array}.$$

For Courses 3 and 4,

$$\begin{aligned} G_3(2) &= 2+6 = 8, \\ G_3(3) &= \max\{2+7, 4+6\} = 10, \\ G_3(4) &= \max\{2+9, 4+7, 7+6\} = 13, \\ G_3(5) &= \max\{2+9, 4+9, 7+7, 8+6\} = 14. \end{aligned}$$

Thus,

$$\begin{array}{c|cccc} s & 2 & 3 & 4 & 5 \\ \hline G_3(s) & 8 & 10 & 13 & 14 \end{array}.$$

For Courses 2–4,

$$\begin{aligned} G_2(3) &= 5 + G_3(2) = 13, \\ G_2(4) &= \max\{5 + G_3(3), 5 + G_3(2)\} = 15, \\ G_2(5) &= \max\{5 + G_3(4), 5 + G_3(3), 6 + G_3(2)\} = 18, \\ G_2(6) &= \max\{5 + G_3(5), 5 + G_3(4), 6 + G_3(3), 9 + G_3(2)\} = 19. \end{aligned}$$

Hence,

$$\begin{array}{c|cccc} s & 3 & 4 & 5 & 6 \\ \hline G_2(s) & 13 & 15 & 18 & 19 \end{array}.$$

Finally, the first-stage calculation and backtracking give

$$\boxed{\begin{aligned} G_1(7) &= \max\{3 + G_2(6), 5 + G_2(5), 6 + G_2(4), 7 + G_2(3)\} \\ &= \max\{22, 23, 21, 20\} = 23, \\ (x_1, x_2, x_3, x_4) &= (2, 1, 3, 1), \qquad \text{maximum grade points} = 5+5+7+6 = 23 \end{aligned}}$$

4. (20 points) A food-truck owner must decide where to operate during each of the next 5 days. Each day, she can choose one of three locations: A, B, and C. The estimated profit from operating at each location on each day is shown below:

Location	Day 1	Day 2	Day 3	Day 4	Day 5
A	8	6	7	9	6
B	5	9	6	7	10
C	6	7	10	5	8

However, changing locations from one day to the next incurs a travel and setup cost. The cost of moving from one location to another is:

From/To	A	B	C
A	0	3	4
B	3	0	2
C	4	2	0

At the beginning of Day 1, the truck is already located at A. Therefore, if she operates at location A on Day 1, there is no travel cost. If she operates at location B or C on Day 1, she must pay the corresponding travel cost from A. There is one additional restriction: she may not operate at the same location for three consecutive days (i.e., she may operate at the same location for at most two consecutive days). The goal is to choose the operating location for each of the 5 days to maximize total profit minus travel costs.

Suppose we define the state before day t as $S_t = (L, r)$, where $L \in \{A, B, C\}$ is the location where the truck operated most recently before day t, and $r \in \{0, 1, 2\}$ is the number of consecutive days, immediately before day t, that the truck has operated at location L. For Day 1, the initial state is $S_1 = (A, 0)$.

Solve this problem using backward induction. Find the optimal sequence of locations and the maximum total net profit (you do not need to write down recursive equations, but you must show the dynamic programming calculations clearly).

Solution.

Let $V_t(L, r)$ be the maximum net profit from Days t through 5 when the state before Day t is (L, r). Each entry below includes the current day's profit, travel cost, and optimal continuation value. A dash denotes an infeasible third consecutive day.

Day 5.

State	A	B	C	V_5/best
$(A, 1)$	6	7	4	$7/B$
$(A, 2)$	$--$	7	4	$7/B$
$(B, 1)$	3	10	6	$10/B$
$(B, 2)$	3	$--$	6	$6/C$
$(C, 1)$	2	8	8	$8/B$ or C
$(C, 2)$	2	8	$--$	$8/B$

Day 4.

State	A	B	C	V_4/best
$(A,1)$	16	14	9	$16/A$
$(A,2)$	$--$	14	9	$14/B$
$(B,1)$	13	13	11	$13/A$ or B
$(B,2)$	13	$--$	11	$13/A$
$(C,1)$	12	15	13	$15/B$
$(C,2)$	12	15	$--$	$15/B$

For example, from $(C,1)$ on Day 4, choosing B gives $7-2+V_5(B,1)=15$.

Day 3.

State	A	B	C	V_3/best
$(A,1)$	21	16	21	$21/A$ or C
$(A,2)$	$--$	16	21	$21/C$
$(B,1)$	20	19	23	$23/C$
$(B,2)$	20	$--$	23	$23/C$
$(C,1)$	19	17	25	$25/C$
$(C,2)$	19	17	$--$	$19/A$

Day 2.

State	A	B	C	V_2/best
$(A,1)$	27	29	28	$29/B$
$(A,2)$	$--$	29	28	$29/B$
$(B,1)$	24	32	30	$32/B$
$(B,2)$	24	$--$	30	$30/C$
$(C,1)$	23	30	26	$30/B$
$(C,2)$	23	30	$--$	$30/B$

Day 1 from $(A,0)$.

	A	B	C	$V_1(A,0)$
Net value	$8+29=37$	$(5-3)+32=34$	$(6-4)+30=32$	37

Backtracking gives the unique optimal sequence (A,B,C,B,B). The daily net contributions are

$$8, \quad 9-3=6, \quad 10-2=8, \quad 7-2=5, \quad 10,$$

so the maximum total net profit is $8+6+8+5+10=37$. Therefore,

$$\boxed{\begin{array}{c}\text{optimal sequence } (A,B,C,B,B),\\ \text{maximum net profit } =37\end{array}}$$

The final two days at B are allowed; only a third consecutive day would violate the restriction.

5. (20 points) Your system consists of a single CPU with finite buffer capacity. Jobs arrive according to a Poisson process with rate λ jobs/sec. The job sizes are exponentially distributed with mean $1/\mu$ seconds. Jobs are serviced in FCFS order. Let $K-1$ denote the maximum number of jobs that your system can hold in the queue. Thus, including the job in service, there are a maximum of K jobs in the system at any one time. If a job arrives when there are already K jobs in the system, then the arriving job is rejected.

Your Defense Advanced Research Projects Agency (DARPA) proposal requires that you reduce the probability of rejecting jobs in your system. To do this you could either ask for money to double the system capacity, or, alternatively, you could ask for money to double the speed of the CPU so that jobs get processed faster, thereby lowering the probability that there are K jobs in the system. Assuming both proposals have the same cost, which do you choose? (Asking for both is too expensive.)

Suppose that $K = 4$, $\lambda = 2$, and $\mu = 4$. Answer the following:

(a) Draw the CTMC.

(b) Derive the stationary probabilities π_n for $n = 0, \ldots, 4$.

(c) What is the utilization of the system, i.e., the fraction of time that the server is busy?

(d) What is the fraction of jobs turned away (loss probability)? Use the word PASTA in your explanation.

(e) What is $\mathbb{E}[N]$, i.e., the expected number of jobs in the system?

(f) What is $\mathbb{E}[T]$ for only those jobs that enter the system? Hint: use the effective arrival rate $\lambda_e := \sum_{n=0}^{K} \lambda_n \pi_n$, where λ_n is the arrival rate to the system with n jobs.

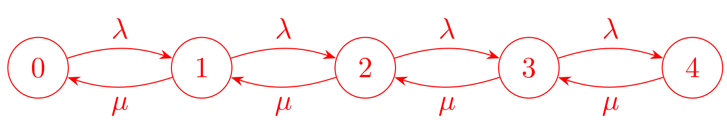
(g) Which has the greater effect on lowering loss probability: doubling the system capacity from $K = 4$ to $K = 8$ or doubling the CPU speed from $\mu = 4$ to $\mu = 8$?

(h) Answer the same question as (g), except now $K = 4$, $\lambda = 3$, and $\mu = 4$.

(i) Explain intuitively why (g) and (h) resulted in different answers.

Solution.

(a) The CTMC is



Arrivals in state 4 are rejected and therefore do not change the state.

(b) Local balance gives

$$\lambda \pi_n = \mu \pi_{n+1}, \qquad n = 0, 1, 2, 3.$$

With $\rho = \lambda/\mu = 1/2$, we have $\pi_n = \rho^n \pi_0$. Normalization gives

$$1 = \pi_0 \sum_{n=0}^{4} \left(\frac{1}{2}\right)^n = \pi_0 \frac{31}{16}.$$

Therefore,

$$\boxed{(\pi_0, \pi_1, \pi_2, \pi_3, \pi_4) = \left(\frac{16}{31}, \frac{8}{31}, \frac{4}{31}, \frac{2}{31}, \frac{1}{31}\right)}$$

(c) The utilization is the probability that the server is busy:

$$1 - \pi_0 = \frac{15}{31} \approx 0.4839.$$

(d) By PASTA (Poisson Arrivals See Time Averages), an arrival sees a full system with the steady-state probability π_4. Hence

$$P_{\text{loss}} = \pi_4 = \frac{1}{31} \approx 0.0323.$$

(e)

$$\mathbb{E}[N] = \sum_{n=0}^{4} n\pi_n = \frac{26}{31} \approx 0.8387.$$

(f) The effective arrival rate of admitted jobs is

$$\lambda_e = \lambda(1 - \pi_4) = 2\left(1 - \frac{1}{31}\right) = \frac{60}{31}.$$

By Little's Law for admitted jobs,

$$\mathbb{E}[T] = \frac{\mathbb{E}[N]}{\lambda_e} = \frac{26/31}{60/31} = \frac{13}{30} \approx 0.4333 \text{ seconds}.$$

(g) For an $M/M/1/K$ queue,

$$\pi_K = \frac{\rho^K}{\sum_{n=0}^{K} \rho^n}.$$

If capacity is doubled to $K = 8$, $\rho = 1/2$ and

$$P_{\text{loss}} = \frac{(1/2)^8}{\sum_{n=0}^{8} (1/2)^n} = \frac{1}{511} \approx 0.00196.$$

If speed is doubled to $\mu = 8$, then $K = 4$, $\rho = 1/4$, and

$$P_{\text{loss}} = \frac{(1/4)^4}{\sum_{n=0}^{4} (1/4)^n} = \frac{1}{341} \approx 0.00293.$$

(g) Doubling system capacity is better: $1/511 < 1/341$.

(h) If capacity is doubled, then $K = 8$ and $\rho = 3/4$:

$$P_{\text{loss}} = \frac{(3/4)^8}{\sum_{n=0}^{8} (3/4)^n} = \frac{6561}{242461} \approx 0.0271.$$

If speed is doubled, then $K = 4$ and $\rho = 3/8$:

$$P_{\text{loss}} = \frac{(3/8)^4}{\sum_{n=0}^{4} (3/8)^n} = \frac{81}{6505} \approx 0.0125.$$

(h) Doubling CPU speed is better: $81/6505 < 6561/242461$.

(i) When $\rho = 1/2$, the system is lightly loaded and seldom fills, so four extra buffer slots nearly eliminate an already rare rejection event. When $\rho = 3/4$, sustained service congestion is the main issue. Extra buffer space only stores waiting jobs, while faster service directly removes the bottleneck and reduces the probability of reaching capacity.

6. (10 points) Suppose your computer center currently consists of a single server of speed μ. Jobs arrive according to a Poisson process with rate λ, and their service times are exponentially distributed.

Suppose the current response time is considered intolerable by the users. A second, faster server, running at speed $\alpha\mu$ (for $\alpha > 1$), is purchased and added to the system. With the second server, the system becomes a variant of the $M/M/2$ queue where the service rates of the two processors are not identical. Denote the service rate of the first processor by $\mu_1 = \mu$ and the service rate of the second processor by $\mu_2 = \alpha\mu$, where $\mu_1 < \mu_2$. In such a system, when both servers are idle, the faster server is scheduled for service before the slower one. Define the utilization, $\rho < 1$, by

$$\rho = \frac{\lambda}{\mu_1 + \mu_2}.$$

Suppose the mean response time is given by

$$\mathbb{E}[T] = \frac{1}{\lambda A(1-\rho)^2},$$

where

$$A = \frac{\mu_1\mu_2(1+2\rho)}{\lambda(\lambda+\mu_1)} + \frac{1}{1-\rho}.$$

Determine the mean number of jobs $\mathbb{E}[N]$ in the system. You do not need to simplify your answer.

Solution.

By Little's Law, $\mathbb{E}[N] = \lambda\mathbb{E}[T]$. Therefore,

$$\mathbb{E}[N] = \lambda\frac{1}{\lambda A(1-\rho)^2} = \frac{1}{A(1-\rho)^2}.$$

Hence,

$$\boxed{\mathbb{E}[N] = \frac{1}{A(1-\rho)^2}}$$

where

$$A = \frac{\mu_1\mu_2(1+2\rho)}{\lambda(\lambda+\mu_1)} + \frac{1}{1-\rho}, \qquad \rho = \frac{\lambda}{\mu_1+\mu_2} < 1.$$

7. (10 points) In Question 6, a hotshot consultant walks in and makes the radical proposal of disconnecting the original server entirely (i.e., simply letting the faster server run by itself). Clearly this makes sense with respect to power, but the consultant claims this is also a win for $\mathbb{E}[N]$. For a consulting fee of \$400/hr, what is the consultant thinking? Come up with an instance, with specific values for λ, μ_2, and μ_1 for which the consultant is right. Also, explain intuitively what is happening. [If you find this problem interesting, you can think about a general criterion under which the consultant would be right. Throwing away servers is fun!]

Solution.

Choose

$$\boxed{\lambda = 5, \qquad \mu_1 = 1, \qquad \mu_2 = 10}$$

Both systems are stable because $5 < 1 + 10$ and $5 < 10$. For the two-server system,

$$\rho = \frac{5}{1+10} = \frac{5}{11},$$

and, using the formula printed in Question 6,

$$\begin{aligned} A &= \frac{(1)(10)(1 + 2 \cdot 5/11)}{5(5+1)} + \frac{1}{1 - 5/11} \\ &= \frac{7}{11} + \frac{11}{6} = \frac{163}{66}. \end{aligned}$$

Therefore,

$$\mathbb{E}[N]_{\text{two-server}} = \frac{1}{(163/66)(1 - 5/11)^2} = \frac{1331}{978} \approx 1.3609.$$

If the slower server is disconnected, the system becomes an $M/M/1$ queue with arrival rate 5 and service rate 10. Its mean number of jobs is

$$\mathbb{E}[N]_{\text{fast-only}} = \frac{\lambda}{\mu_2 - \lambda} = \frac{5}{10 - 5} = 1.$$

Thus,

$$\boxed{\mathbb{E}[N]_{\text{fast-only}} = 1 < \frac{1331}{978} = \mathbb{E}[N]_{\text{two-server}}}$$

Disconnecting the slower server reduces the mean number of jobs in this instance.

Intuition: Under the fast-server-first rule, an arrival that finds the fast server busy but the slow server idle is sent directly to the slow server. With the large speed difference here, that job may spend much longer in slow service than it would spend waiting briefly for the fast server and then receiving fast service. The fast server alone has utilization $5/10 = 1/2$, so it has enough spare capacity for this waiting-for-fast strategy to reduce $\mathbb{E}[N]$.